GestureHeal: Zero-Hardware Neurological Rehabilitation Using Computer Vision and Adaptive Gamification

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***Abstract*—Neurological rehabilitative medicine is crucial for restoring motor skills and improving the quality of life for people with neurological disorders like strokes, Parkinson's Disease, and traumatic brain injuries. Current rehabilitative technologies for these populations generally rely on expensive hardware-based tools, require specialized clinical environments, and require ongoing direct supervision from trained professionals, making them difficult to access and adhere to over the long term. Recently, developments in computer vision and gamification have enabled the creation of interactive rehabilitative technologies that deliver rehabilitation services remotely via gaming, using only a standard webcam. GestureHeal is a rehabilitation system that uses real-time motion estimation, adaptable gamification logic, and real-time motion analysis to provide interactive rehabilitation services with no special hardware required for users.**

The GestureHeal system uses MediaPipe for real-time motion tracking of the user's body to provide tracking data as well as interfaces for games developed for use in a rehabilitation setting and analytics about how well users performed while completing their rehabilitation exercises (including capability for both clinicians and patients) and how much rehabilitation activity they did. A proof of concept prototype was constructed that demonstrates the feasibility of the system and includes four core components: (1) a means for onboarding patients; (2) a means for providing real-time tracking of motion using gesture-based identification of movement, (3) a means for tracking user movements for gaming to conduct gamified rehabilitation exercises, and (4) an automatic generation of a clinical summary report for the clinician. Future Work may include the clinical validation of intelligent rehabilitation methods. GestureHeal has the potential to become an accessible, inexpensive, scalable platform for home-based neurological rehabilitation.

***Index Terms*—Neurological Rehabilitation, Computer Vision, Human Pose Estimation, Gamification, Tele-Rehabilitation, MediaPipe, Healthcare AI.**

I. Introduction

Neurological diseases often precede chronic disabilities. Chronic disabilities can emerge from the acute effects of these neurological disorders (e.g., stroke, Parkinson's syndrome, traumatic brain injury, musculoskeletal disorders) [21]. To support motor recovery and functional independence after a neurological impairment, rehabilitation requires frequent, individualized, and lengthy therapeutic programs. However, traditional rehabilitation methods are frequently limited by their reliance on therapists and specialized equipment and require regular visits to rehabilitation facilities, resulting in high costs and limited access for many people, especially in rural locations.

New developments in human-computer interaction (HCI) and markerless pose estimation have opened new ways to provide rehabilitation services in the home. MediaPipe [19], OpenPose, and vision-based motion analysis systems such as BlazePose [20] can deliver body kinematic information from a conventional camera without the need for wearable sensors or specialized hardware. The development of gamification techniques to improve patient motivation and exercise adherence through the use of an interactive approach to repeatable rehabilitation exercises also shows promise.

The majority of rehabilitation systems on the market today use expensive sensors, robotics, or virtual reality equipment that are not amenable to large-scale implementations in homes [4]. The most prevalent rehabilitation solution also tends to isolate different types of functionality into related categories (motion tracking, gamification, or tele-monitoring) without providing an integrated and adaptive rehabilitation ecosystem.

To overcome these barriers, this paper describes a new neurological rehabilitation solution called GestureHeal. GestureHeal is a zero-hardware neurological rehabilitation solution that contains real-time pose recognition, gamified rehabilitation activities, automated movement evaluations, and remote monitoring between therapists and patients within a unified framework. GestureHeal uses a commercially available webcam and browser-based methods for creating computer vision to provide affordable and accessible rehabilitation experiences through low-cost implementation.

Key contributions include:

- Introduction of a zero-hardware rehabilitation system, GestureHeal, that takes advantage of the capabilities of browser-based computer vision, adaptive gaming and automated movement analysis for treatment of neurologic disorders at home.

- The architecture of the system involves modular implementation of real-time video-based pose estimation, calculation of rehabilitation metrics, and interactive gaming using off-the-shelf components. ,
- This work highlights the technical ability to deliver neurological rehabilitation through the use of browser technology without the use of motion trackers, inclusive of hardware components.
- Future directions for multimodal, intelligent, and clinically validated rehabilitation systems include future research that will explore the possibility of making the rehabilitation process adaptive and giving the clinician analytic insights into the patient's recovery.

II. Related Work

A. Existing Rehabilitation Technologies

The objectives of neurorehabilitation comprise motor function recovery and reduction in disabilities, as well as an improvement in the quality of life [3], [8], [15]. Although physical therapy involves repetitive tasks performed by expert therapists, it possesses numerous weaknesses linked to therapist specialization, the duration of therapy sessions, dependency on exercise compliance at home, and inefficiency in objectively assessing patient performance [2]. Exercise therapy and other modalities such as manual therapy often lack an effective, quantitative measure of patient progress [4].

Rehabilitation technologies include several technological advances that have both benefits and disadvantages [12], [17]. Sensor- and robotic-based assessment and therapeutic devices have the benefit of high accuracy, but suffer from very high costs (mainly hardware) and only have a use in a clinical environment [17]. Virtual reality (VR)/augmented reality (AR)/mixed reality (MR) technologies provide patients with context-specific learning and increase motivation; however, they are difficult to use in the home setting due to the requirement for a head-mounted device [8], [18] and have uncertain effectiveness as shown in clinical studies [23]. Other types of technologies that do not require physical markers (i.e. MediaPipe, OpenPose, YOLO-Pose) allow for the assessment of postures and movements without the use of physical markers [1], [14], but the trade-off between the accuracy of the estimate of the posture/movement, the ability to provide an output in real-time, and the generalizability of the system continues to be investigated [14]. Tele-rehabilitation and gamification-based approaches provide feedback to patients or caregivers about their progress and increase motivation [12], [22].

While there have been successful implementations, the limitations still include: high cost of equipment and software, lack of individualized treatment plans, limited availability of real-time feedback, and limited ability for independent home use of physical therapy [12], [17]; General. The suggested solution, GestureHeal, was developed to alleviate all of these constraints through the use of inexpensive web cams to monitor patients' posture in real time, provide corrective feedback through the use of WebAR skeleton overlays, provide adaptive gaming features, and provide automatic generation and delivery of clinical reports based on facial recognition for measuring pain.

B. Comparative Analysis of Existing Rehabilitation Systems

Table I presents a tally of the technology category most central to each directly classifiable primary study cited in this paper's reference list (Refs. [1], [13], [18]; review-type references [30], [17], [24] are excluded from the count as they span multiple categories). Fig. 1 visualizes this distribution. Consistent with the qualitative findings reported in recent systematic reviews [12], [24], virtual reality is the most frequently represented category among the cited primary studies, while markerless computer-vision-based methods – the approach adopted by GestureHeal – remain comparatively underexplored despite their low hardware footprint [1], [14]. Table II further presents a feature-level comparison of representative individual systems, highlighting differences in hardware requirements, home deployment capability, gamification support, and personalization. Although there are many different interactive rehabilitation technologies, the majority of them rely on either specialized equipment or do not have personalization and progress monitoring available as features in their designs.

This tendency to focus on VR-based rehabilitation frameworks in the above mentioned literature is understandable, since most VR technologies require some type of head-mounted display or additional equipment (tracking devices or similar), which increases costs associated with that design and makes implementation in a home setting (especially by the elderly and persons with mobility impairments) impractical [18].

On the other hand, only a small number of studies have looked at markerless computer vision systems for rehabilitation, indicating there are many potential uses for webcams, these types of studies have not been widely conducted. It is this gap in existing literature (academic, practitioner, and technical) that necessitates the development of a framework such as GestureHeal to address accessibility concerns rather than immersive and/or precise haptic capabilities of the rehabilitation technology itself.

TABLE I: Distributions of the Cited Primary Research Articles According to Technology Type

Technology Type	**Cited Refs.**	**Count**
Virtual Reality	[5],[8],[9],[13],[18]	5
Augmented Reality	[6],[7]	2
Robotic / Sensor-Based	[4],[11]	2
Computer Vision / Pose Est.	[1],[10]	2

TABLE II: Comparison of Existing Rehabilitation Systems

Study	**Game**	**Home**	**Hardware**	**Personal.**
[1]	Yes	Yes	Low	No
[5]	Yes	Limited	High	Partial
[6]	Yes	Yes	Medium	No
[10]	Yes	Yes	Medium	No
GestureHeal	Yes	Yes	Very Low	Proposed

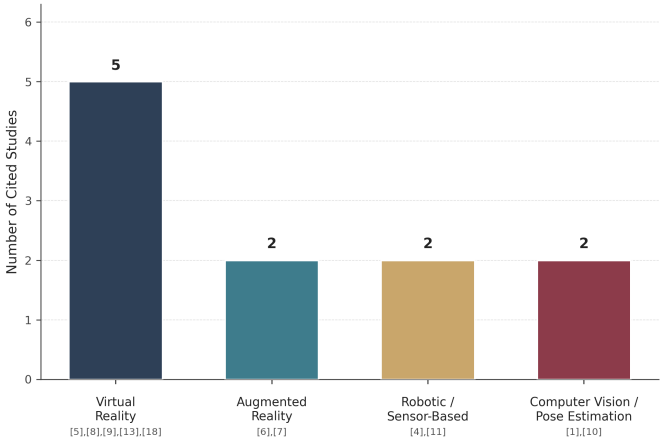


Fig. 1: Distribution of cited primary studies by technology category, based on a direct tally of References [1], [4]–[11], [13], [18] (n = 11 directly classifiable primary studies).

III. Research Gap and Identified Challenges

The current research literature shows significant advancement in the following fields: computer vision-assisted rehabilitation, virtual reality rehabilitation, and gamification rehabilitation systems [12], [17]. However, several challenges remain that do not allow widespread usage of these.

To begin with, the majority of rehab platforms rely on specialized hardware (robotics, sensor technology, cameras, or head-mounted displays), leading to high cost and restricting consumers from using these types of systems in their home [4], [17]. Most of the vision rehabilitation systems are more focused on movement observation and evaluation rather than incorporating the patient into therapy with their adaptive progress and continuous observation [14]. There are also countless gamified rehabilitation systems available that do not provide any individualized exercise plans or a clinician-facing interface [11], [14]. Lastly, there is no one rehabilitation framework that supports real-time pose detection, automatic movement evaluation, gamification, remote observation, and further scalability to intelligent personalization and management of pain.

These limitations, summarized qualitatively in Table III, are dominated in the literature's discussion by hardware dependency and limited personalization, motivating the need for an accessible, low-cost, and integrated rehabilitation platform capable of delivering engaging home-based therapy while providing a foundation for future intelligent rehabilitation technologies. GestureHeal addresses this gap by proposing a zero-hardware rehabilitation ecosystem that combines computer vision, gamification, and remote monitoring within a single scalable framework.

As summarized in Section IV-B, these challenges are interconnected and compound one another in practice, motivating a holistic design approach rather than incremental improvements to existing hardware-dependent systems.

TABLE III: Challenges in Existing Rehabilitation Systems

Challenge	**Impact**
Hardware Dependency	Reduced accessibility and increased cost
Limited Personalization	Reduced patient engagement
Lack of Monitoring	Difficult long-term assessment
Fixed Exercise Protocols	Low adaptability
Limited Accessibility	Reduced scalability

The present system lacks an efficient, inexpensive system that combines computer vision for the web, game adaptation, and remote monitoring functions. GestureHeal meets this need with a comprehensive no-hardware rehabilitation system.

IV. Proposed GestureHeal Framework

GestureHeal is being designed as a zero-hardware neurological rehabilitation system that utilizes an integrated system of real-time computer vision, gamification, and automated analysis of movement in one web-based platform. It is designed for individuals with a variety of diseases affecting the nervous or musculoskeletal systems, including stroke, Parkinson's disease, traumatic brain injury (TBI), cervical disorders, and impairments to the upper extremities. The following are design objectives for the system:

- To develop a gamified rehabilitation system for use by patients within their homes.
- To monitor the current level of pain via a patient's facial expressions in real-time
- To automatically adjust the challenge level of the task based on the patient's performance level
- To provide a dashboard for tracking and alerting clinicians
- To require only a camera as the hardware necessary
- To positively affect the long-term compliance and recovery of patients
- To develop a VR Skeleton overlay using WebAR, producing a real image of the patient's body with a camera.

A. Methodology Workflow

As depicted in Fig. 2, the workflow of the GestureHeal system is composed of the following major components: patient onboarding (where patients describe the conditions upon which to rehab, along with identified locations on their body where they would like to receive rehabilitation), an AI-based calibration phase to determine the patient's body posture via a conventional webcam, a continuous utilization of video footage to derive measurable metrics through the MediaPipe framework (including range of motion, bilateral symmetry, movement-smoothness, and tremor-type measurements for rehabilitation), and the creation of a gamified rehabilitation exercise program designed to turn traditionally monotonous, repetitive rehabilitation activities into entertaining and engaging exercises. The overall GestureHeal framework also has plans to include additional features (Session Analytics and Automatic Reporting Systems) for tracking and/or monitoring patient progress over time, while there are plans to set up Remote Patient Monitoring and feedback from Clinicians in the future.

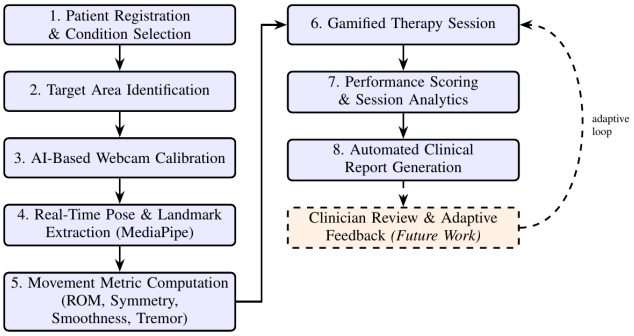


Fig. 2: Methodology workflow of the proposed GestureHeal framework.

B. System Architecture

The overall system architecture of GestureHeal is shown in Fig. 3. The framework consists of a webcam-based client/presentation layer, a computer vision layer responsible for pose estimation and motion analysis, an application/backend layer providing REST services and authentication, a persistent data layer, and an analytics and reporting layer that generates session summaries and clinical reports. The modular, layered architecture facilitates future integration of intelligent rehabilitation features – including facial pain detection, WebAR guidance, and clinician-oriented monitoring – without requiring redesign of the existing pipeline.

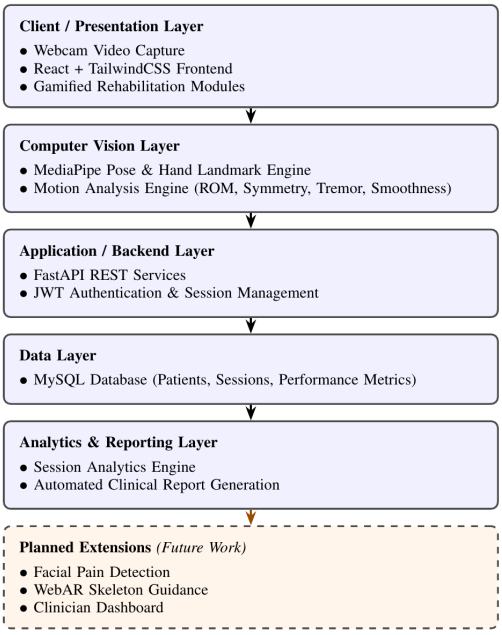


Fig. 3: System architecture of the proposed GestureHeal framework.

C. Functional Modules and Rehabilitation Metrics

The Table IV of metrics that comprise the metrics table on rehabilitation, including some regarding neurologically and/or musculoskeletally based processes, are calculated from MediaPipe skeletal landmarks and are updated during every session so that real-time feedback may be accessed during a given session, and for possibly the longer term via subsequent annual comparisons of pre- and post-evaluations achieved with the same process Table V.

TABLE IV: Rehabilitation Metrics Computed by GestureHeal

Metric	**Purpose**
Range of Motion	Joint mobility assessment
Bilateral Symmetry	Recovery monitoring
Tremor Index	Motor stability analysis
Movement Smoothness	Coordination assessment

The above framework calculates and evaluates rehabilitation-related metrics as defined by best practice in clinical care, utilizing a real-time pose estimation (per frame of time) process and geometric (3D) analysis of skeletal landmarks. In light of the above, metrics obtained from the above will serve to address movement-related (not diagnostic) metrics related to rehabilitation. The results indicate that real-time pose estimation can be accomplished using commercially available consumer cameras (with and without calibration). Future research will focus on the validity of the metrics produced against the gold standard of clinical practice, as well as longitudinal studies (over time) with patient populations.

TABLE V: Clinical Rehabilitation Objectives Supported by GestureHeal

Condition	**Goal**	**Exercise**
Stroke	Improve ROM	Arm Raise
Parkinson's	Tremor Reduction	Hand Tracking
TBI	Coordination Recovery	Reach Exercise
Cervical	Improve Flexibility	Neck Rotation
Shoulder	Mobility Improvement	Shoulder Abduction

V. Prototype Implementation

The development of the GestureHeal prototype allowed for an understanding of the feasibility of vision-based rehabilitation without the need for specialized hardware. The prototype will track motion in real time using the MediaPipe Pose and Hand tracking modules through a camera without any markers on the body to determine where the body is located. The extracted landmarks are processed to derive rehabilitation-related metrics including joint angles, movement trajectories, and upper-limb motion characteristics. To improve patient engagement, the prototype incorporates an interactive gamified rehabilitation environment in which body movements are directly mapped to gameplay mechanics; a rehabilitation-oriented Fruit Ninja module has been implemented to validate the integration between real-time pose estimation and therapeutic exercises. Additional features implemented in the software include patient registration, condition selection, session management, and automated report generation.

A. Implementation Requirements

The prototype of the application was developed and tested on a standard laptop-class development machine (Intel i5/Ryzen 5 class processor, 8 GB RAM, ˜10 GB free storage) with a minimum 720p webcam, whereas the target patient-side application requires only an entry-level computer (Intel i3 class,4 GB RAM) with a built-in or external webcam supporting Windows, macOS, or Linux OS. As far as the software stack is concerned, Python and JavaScript were used with NumPy,Pandas, and Matplotlib for data processing; React.js, TailwindCSS, and Framer Motion for the frontend; MediaPipePose, OpenCV, TensorFlow/TF.js, and Scikit-learn for AI and computer vision; HTML5 Canvas and Phaser.js for game rehabilitation; Node.js/Express.js (with Flask/Django as alternatives) for backend services; and Recharts, jsPDF, and html2 canvas for reports.The system is further designed to satisfy key non-functional requirements: real-time performance at approximately 30 FPS, end-to-end response latency below 100 ms, an elderly-friendly user interface, secure handling of patient data, and cross-platform compatibility.

B. System Deployment Specifications

For ease of deployment and accessibility, the GestureHeal prototype is developed through the usage of up-to-date web technology as outlined in Table VI.

TABLE VI: System Deployment Specifications

Component	**Technology**
Frontend	React + TailwindCSS
Pose Estimation	MediaPipe Pose and Hands
Backend	FastAPI
Database	MySQL
Authentication	JWT
Deployment	Vercel + Railway
Camera Requirement	Standard Webcam
Browser Compatibility	Chrome, Edge, Firefox

The backend architecture will store the users' data, therapy sessions, and their performance metrics, thus making longitudinal studies possible in the future. The present implementation demonstrates the technical feasibility of delivering low-cost rehabilitation using only a webcam and commodity computing devices, while also providing a foundation for future intelligent rehabilitation modules.

VI. Future Enhancements

The adaptive feedback enabled by the gamification component of GestureHeal is intended to support a transition from traditional, passive physical therapy toward an interactive therapy practice. The prototype's adaptive feature of adjusting difficulty, rewards, and hints based on metrics will encourage users to complete the rehabilitation session and will increase completion and retention rates among the users. The use of a standard webcam instead of special hardware will significantly reduce the overall costs of rehabilitation for both the service provider and the client, which will create greater access to rehabilitation for low-income or rural populations.

Additionally, the use of no special equipment allows for both clinician-monitored and freestanding rehabilitation modes in the user's home.

A. Clinical Validation

The subsequent step will include controlled and longitudinal studies to assess GestureHeal's usability, effectiveness, and safety across various populations with the cooperation of healthcare professionals.

B. Multimodal Rehabilitation

Moreover, in addition to the analysis of facial expressions, the use of motion tracking via webcam together with additional modalities (inertial, electromyographic, cardiopulmonary, and voice) enables the capturing of more precise recovery dynamics that cannot be achieved by the use of motion tracking in one modality alone. The facial pain detection component is envisioned to estimate pain-related facial expressions such as grimacing, while future work will evaluate robustness across varying lighting conditions and diverse populations. In combination with performance scoring, each patient's score will be adjusted based on either self-reported or detected pain level via comparison of performance score vs pain level. The proposed scoring mechanism aims to make the rehabilitation process sensitive to patient discomfort. The entire system will use an Adaptive Difficulty Engine and WebAR Skeleton Guidance to form a closed-loop personalized rehabilitation system. Further phases will investigate the use of performance trajectories and reinforcement learning-based policies to proactively adjust task difficulty of tasks while using spatialized visual overlays to assist in correcting kinematics without the use of additional hardware.

VII. Limitations

Although GestureHeal has shown evidence of technical feasibility, there continue to be some limitations. The present work represents a prototype and conceptual framework and has not yet undergone clinical validation or large-scale user studies. The proposed metrics are intended to demonstrate feasibility rather than establish diagnostic accuracy. No formal clinical trials of the system have yet been conducted, and the platform has not been evaluated on large, diverse patient datasets. Additional technical limitations include sensitivity of monocular RGB pose estimation to poor lighting conditions, occlusion, and camera placement; the current absence of haptic or depth-sensing feedback; and the need for further evaluation of landmark-detection accuracy across diverse body types and skin tones. Further studies involving patient cohorts and longitudinal evaluation are necessary. These limitations do not diminish the contribution of the proposed framework, but rather define a clear validation roadmap for GestureHeal.This work is presented as a prototype-driven feasibility study and conceptual research framework intended to stimulate future investigations into clinically validated zero-hardware rehabilitation systems.

VIII. Conclusion

GestureHeal presents a zero-hardware neuro-rehabilitation system that combines real-time pose estimation, adaptive gamification, automated movement evaluation, and home exercise tracking to deliver clinically inspired rehabilitation exercises while addressing key limitations of existing rehabilitation systems. Its significance lies in the potential to transform rehabilitation delivery by making it accessible, cost-effective, and motivating for patients, while supporting continuity of care – an aspect that is especially important in resource-constrained settings. Future work will focus on pain-sensitive adaptability, intelligent personalization, WebAR-based guidance, clinician-focused analytics, multimodal evaluation, and, most importantly, large-scale deployment and validation through clinical trials. The present work should therefore be interpreted as a feasibility and systems study rather than a clinically validated therapeutic intervention.GestureHeal is intended to provide a research framework that may inform the development of future intelligent rehabilitation systems. This study's purpose is not to act as a therapeutically validated clinical application but to serve as a foundation for research and feasibility studies. The described framework is not meant to substitute current rehabilitation techniques but only to add to the current treatment model with affordable and easy-to-use technology.

References

[1] K. E. Ch. Vidyasagar et al., "Gamified rehabilitation and physiotherapy using MediaPipe for posture and hand gesture recognition," 2024.

[2] I. Martínez et al., "Effectiveness of a gamified and home-based approach for upper-limb rehabilitation," 2022.

[3] Y. Wang et al., "Post-stroke resting-state EEG connectivity: A longitudinal neuro-rehabilitation study," 2022.

[4] L. Rossi et al., "A hybrid approach to neurorehabilitation: Validating an integrated EEG-EMG/robotic system for wrist therapy," 2023.

[5] A. Bouatrous et al., "A virtual reality-based serious game designed for personalized hand motor rehabilitation," 2023.

[6] M. Gaspar et al., "Augmented reality serious games for smart physical rehabilitation," 2023.

[7] M. Taufiq et al., "Designing an Augmented Reality Based Serious Game for Rehabilitation of Post Stroke Patient's Hand Coordination and Motor Skills," 2023.

[8] P. Domínguez-Tellez, J. A. Moral-Munoz, A. Salazar, E. Casado-Fernandez, and D. Lucena-Anton, "Game-based virtual reality interventions to improve upper limb motor function and quality of life after stroke: Systematic review and meta-analysis," *Games Health J.*, vol. 9, no. 1, pp. 1–10, 2020.

[9] A. Baldominos, Y. Saez, and C. G. D. Pozo, "An approach to physical rehabilitation using state-of-the-art virtual reality and motion tracking technologies," *Proc. Comput. Sci.*, vol. 64, pp. 10–16, Jan. 2015.

[10] R. Baluz, A. Teles, J. E. Fontenele, R. Moreira, R. Fialho, P. Azevedo, D. Sousa, F. Santos, V. H. Bastos, and S. Teixeira, "Motor rehabilitation of upper limbs using a gesture-based serious game: Evaluation of usability and user experience," *Games Health J.*, vol. 11, no. 3, pp. 177–185, Jun. 2022.

[11] Sharafianaradakani et al., "A Web-Based Gamification of Upper Extremity Robotic Rehabilitation," IEEE, 2023.

[12] J. J. Sanchez-Gil, A. Sáez-Manzano, R. López-Luque, J.-J. Ochoa-Sepúlveda, and E. Cánete-Carmona, "Gamified devices for stroke rehabilitation: A systematic review," *Comput. Methods Programs Biomed.*, vol. 258, art. no. 108476, 2025.

[13] P. Rodrigues, C. Quaresma, M. Costa, F. Luz, and M. M. Fonseca, "Virtual Reality-Based Telerehabilitation for Upper Limb Recovery Post-Stroke: Systematic Review of Design Principles, Monitoring, Safety, and Engagement Strategies," *arXiv preprint arXiv:2501.06899*, 2025. [Online]. Available:

[14] D. Robinson, A. Gupta, R. Quershi, Q. Fu, and M. Shah, "STROKEVISION-BENCH: A Multimodal Video and 2D Pose Benchmark for Tracking Stroke Recovery," *arXiv preprint arXiv:2509.07994*, 2025. [Online].

[15] S. Mugisha, M. Job, M. Zoppi, M. Testa, and R. Molfino, "Computer-Mediated Therapies for Stroke Rehabilitation: A Systematic Review and Meta-Analysis," *Journal of Stroke and Cerebrovascular Diseases*, vol. 31, no. 6, p. 106454, Jun. 2022, doi: 10.1016/j.jstrokecerebrovasdis.2022.106454.

[16] V. Li, N. Kamalakannan, A. Parnandi, H. Schambra, and C. Fernandez-Granda, "The Potential and Limitations of Vision-Language Models for Human Motion Understanding: A Case Study in Data-Driven Stroke Rehabilitation," *arXiv preprint arXiv:2511.17727*, 2025. [Online].

[17] J. J. Sánchez-Gil, A. Sáez Manzano, J. J. Ochoa-Sepúlveda, L. Muñoz-Millán, D. Cáceres-Gómez, R. López-Luque, and E. Cañete-Carmona, "Hardware and Software Design Trends and Considerations in Gamified Stroke Rehabilitation: Results from a Systematic Review," *Medical & Biological Engineering & Computing*, vol. 64, no. 4, pp. 1203–1230, Apr. 2026, doi: 10.1007/s11517-026-03518-y.

[18] B. Cieślik, "Virtual Reality Exergaming in Outpatient Stroke Rehabilitation: A Scoping Review and Clinician Roadmap," *Journal of Clinical Medicine*, vol. 14, no. 20, Art. no. 7227, 2025, doi: 10.3390/jcm14207227.

[19] C. Lugaresi et al., "MediaPipe: A Framework for Building Perception Pipelines," arXiv preprint arXiv:1906.08172, 2019.

[20] V. Bazarevsky et al., "BlazePose: On-device Real-time Body Pose Tracking," *Proc. CVPR Workshops*, 2020.

[21] World Health Organization, "World Report on Disability," WHO Press, Geneva, Switzerland, 2011.

[22] D. Cramer et al., "Telerehabilitation in Neurological Disorders: A Systematic Review and Meta-analysis," *Journal of Telemedicine and Telecare*, vol. 25, no. 10, pp. 639–650, 2019.

[23] R. L. Proffitt and C. Lange, "Considerations in the Efficacy and Effectiveness of Virtual Reality Interventions for Stroke Rehabilitation," *NeuroRehabilitation*, vol. 44, no. 3, pp. 301–308, 2019.

[24] F. Zanatta, A. Giardini, A. Pierobon, M. D'Addario, and P. Steca, "A systematic review on the usability of robotic and virtual reality devices in neuromotor rehabilitation: Patients' and healthcare professionals' perspective," *BMC Health Serv. Res.*, vol. 22, art. no. 523, 2022.

[25] S. Balasubramanian, A. Melendez-Calderon, and E. Burdet, "On the analysis of movement smoothness," *Journal of NeuroEngineering and Rehabilitation*, vol. 12, no. 1, pp. 1–11, 2015.

[26] S. Balasubramanian, A. Melendez-Calderon, and E. Burdet, "Is movement smoothness a useful measure of motor control?" *Frontiers in Neurology*, vol. 3, Art. no. 30, 2012.

[27] B. Kanko, E. Laende, E. Davis, and K. Selbie, "Concurrent assessment of gait kinematics using marker-based and markerless motion capture," *Journal of Biomechanics*, vol. 127, Art. no. 110665, 2021.

[28] A. Colyer, H. Evans, D. Cosker, and A. Salo, "A review of the evolution of vision-based motion analysis and the integration of advanced computer vision methods towards developing a markerless system," *Sports Medicine - Open*, vol. 4, no. 1, pp. 1–15, 2018.

[29] S. Prvu Bettger and M. P. Resnik, "Telerehabilitation in the age of COVID-19: An opportunity for learning health system research," *Physical Therapy*, vol. 100, no. 11, pp. 1913–1916, 2020.

[30] R. Laver, S. Adey-Wakeling, M. Crotty, N. Lannin, J. George, and M. Sherrington, "Telerehabilitation services for stroke," *Cochrane Database of Systematic Reviews*, no. 1, Art. no. CD010255, 2020.